DBMS Revision Notes

Unit 1: Introduction to Databases

What is a DBMS?

A **Database Management System (DBMS)** is software that manages, stores, retrieves, and manipulates data efficiently.

> "A DBMS serves as an interface between the user and the database."

Key Characteristics

→ **Data Independence** --- Separation of data and programs

→ **Data Integrity** --- Accuracy and consistency of data

→ **Security** --- Access control mechanisms

→ **Concurrency Control** --- Multiple users simultaneous access

→ **Backup & Recovery** --- Crash recovery mechanisms

Three-Schema Architecture

1. **Physical Level** --- How data is stored (blocks, pages)
2. **Conceptual Level** --- What data is stored (tables, relationships)
3. **External Level** --- How users view data (views)

Important Terminologies

Term	Definition
Tuple	A single row in a table
Attribute	A column in a table
Domain	Set of allowed values for an attribute
Cardinality	Number of tuples in a relation
Degree	Number of attributes in a relation

Unit 2: Relational Model

Relational Algebra Operations

1. **Select (σ)** --- Filter rows `σsalary > 50000(Employee)`
2. **Project (π)** --- Select columns `πname, salary(Employee)`
3. **Union (∪)** --- Combine two relations
4. **Set Difference (-)** --- Tuples in one but not other
5. **Cartesian Product (X)** --- Combine all pairs

Additional Operations:

→ **Join (⋈)** --- Combine related tuples

→ **Division (÷)** --- "All" queries

SQL Example

```
SELECT e.name, d.dept_name
FROM Employee e
JOIN Department d ON e.dept_id = d.dept_id
WHERE e.salary > 50000
ORDER BY e.name;
```

Unit 3: Normalization

Functional Dependencies

Definition: X → Y means "X functionally determines Y"

Normal Forms

1. **1NF** --- Atomic values only
2. **2NF** --- 1NF + No partial dependency on candidate key
3. **3NF** --- 2NF + No transitive dependency
4. **BCNF** --- For every FD X → Y, X must be a super key

> **"The key, the whole key, and nothing but the key"** --- 3NF in a nutshell.

Important Exam Tips

→ Always draw **ER diagrams** for conceptual questions

→ Practice writing **SQL queries** with joins and subqueries

→ Know **ACID properties** --- Atomicity, Consistency, Isolation, Durability

→ Understand **serializability** and **conflict equivalence**

→ Remember the difference between **primary key** and **candidate key**